

6.1 Group 2 (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	6. Inorganic Chemistry (A Level Only)
Topic	6.1 Group 2 (A Level Only)
Difficulty	Hard

Time allowed: 50

Score: /41

Percentage: /100

Question 1a

This question is about magnesium.

Magnesium forms a nitrate, $\text{Mg}(\text{NO}_3)_2$ which decomposes on heating as shown below:



Using oxidation numbers, explain why the reaction involves both oxidation and reduction.

[3 marks]

Question 1b

The values for the first, second and third ionisation energies of magnesium are shown in Table 1.1.

Table 1.1

	Ionisation energy / kJ mol^{-1}
First ionisation energy	738
Second ionisation energy	1451
Third ionisation energy	7732

i)

Write an equation for the second ionisation energy of magnesium.

[2]

ii)

Explain the trend in the first three ionisation energies of magnesium.

[3]

iii)

Explain why magnesium has a greater second ionisation energy than barium.

[3]

[8 marks]

Question 1c

Metal peroxides contain the O-O^- ion.

The peroxides of the Group 2 elements, MO_2 , decompose in a similar way to Group 2 metal carbonates.

i)

Write an equation for the thermal decomposition of strontium peroxide, SrO_2 .

[1]

ii)

Suggest how the temperature at which thermal decomposition of MO_2 occurs varies down Group 2. Explain your answer.

[3]

[4 marks]

Question 2a

Table 2.1 below shows the solubility data for the Group 2 metal hydroxides.

Table 2.1

Group 2 metal hydroxide	Solubility / g dm ⁻³ at 20 °C
Magnesium hydroxide	0.140
Calcium hydroxide	1.730
Strontium hydroxide	17.70
Barium hydroxide	38.90

State and explain the factors responsible for the trend in the solubility of the Group 2 hydroxides.

[2 marks]**Question 2b**

Milk of magnesia, Mg(OH)₂ is used to neutralise stomach acid.

Explain why Mg(OH)₂ **cannot** be used to test for carbon dioxide but Ba(OH)₂ **can**.

[2 marks]

Question 2c

i)

Write an expression for the solubility product of $\text{Sr}(\text{OH})_2$ $K_{\text{sp}} =$

[1]

ii)

Use the data from Table 3.1 in part (a) to calculate the value of K_{sp} at 293 K. Include units in your answer and show your working. $K_{\text{sp}} = \dots\dots\dots \text{units} = \dots\dots\dots$

[2]

[3 marks]

Question 2dUse the data from Table 3.1 in part (a) to calculate the pH of a saturated solution of $\text{Mg}(\text{OH})_2$ at 293 K. Show your working. K_{w} at 293 K = $0.681 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$

pH =

[5 marks]

Question 3a

Calcium carbonate can be thermally decomposed to make calcium oxide.



Table 3.1 below shows the standard entropies of each substance.

Table 3.1

Substance	$\text{CaCO}_3(\text{s})$	$\text{CaO}(\text{s})$	$\text{CO}_2(\text{g})$
$S / \text{JK}^{-1} \text{mol}^{-1}$	89	40	214

- i)
Use the information in Table 3.1 to show that calcium carbonate is stable at room temperature (25 °C).

[4]

- ii)
Calculate the minimum temperature needed to decompose calcium carbonate.

[2]

[6 marks]

Question 3b

Describe and explain the trend that is observed in the thermal decomposition of the Group 2 carbonates.

[3 marks]

Question 3c

i)

Write an equation to show the equilibrium for the solubility product for CaCO_3 . Include state symbols.

[1]

ii)

With reference to your equation in part (c)(i), suggest what is observed when a few cm^3 of concentrated $\text{K}_2\text{CO}_3(\text{aq})$ are added to a saturated solution of CaCO_3 . Explain your answer.

[2]

[3 marks]

Question 3d

CaCO_3 has a solubility product, K_{sp} in water at 298 K of $5 \times 10^{-9} \text{ mol}^2 \text{ dm}^{-6}$

Calculate the solubility of CaCO_3 in water at 298 K, in g dm^{-3}

solubility of CaCO_3 = g dm^{-3}

[2 marks]

Model Answers: Hard

1a

a) This reaction involves both oxidation and reduction because:

- Nitrogen / N goes from +5 to +4; [1 mark]
- Oxygen / O goes from -2 to 0; [1 mark]
- Nitrogen / N is reduced

AND

Oxygen / O is oxidised; [1 mark]

[Total: 3 marks]

- If the question tells you to refer to oxidation numbers, this must be the evidence for whether a substance has been reduced or oxidised
 - You cannot just state whether the species has lost or gained electrons.
- Make sure you learn the rules for oxidation numbers of each element
 - Oxygen has an oxidation number of -2 except in peroxides
 - Magnesium is a Group 2 element so will have an oxidation number of +2
- **Careful:** Don't forget to consider the brackets to check the number of atoms overall

1b

b)

i) The equation for the second ionisation energy is:



- Equation correct; [1 mark]
- State symbols correct; [1 mark]

ii) The trend in ionisation energies of magnesium:

- Ionisation energies increase; [1 mark]
- The second ionisation energy is higher than the first as the nuclear charge is

higher

AND

There is similar shielding; [1 mark]

- There is a large increase between the second and third ionisation energy as the third electron is being removed from the 2p orbital as opposed to the 3s orbital which has greater shielding; [1 mark]

iii) The second ionisation energy of magnesium is greater than barium because:

- Mg^+ has a smaller (ionic) radius

OR

Mg^+ has fewer / less shells; [1 mark]

- There is less shielding; [1 mark]
- (So,) there is more / greater nuclear attraction

OR

(So,) the outer electrons are attracted more strongly (to the nucleus); [1 mark]

[Total: 8 marks]

- You must be able to compare ionisation energies for the same element and different elements in the same group
- For successive ionisation energies of the same element, write out the electronic configurations to help with your explanations of the increase
 - E.g Magnesium is $1s^2 2s^2 2p^6 3s^2$
 - From this, you can clearly see that the third electron will be lost from the 2p orbital as opposed to the 3s orbital
 - Hence, a large amount of energy is needed to remove it

1c

c)

i) The equation for the thermal decomposition of SrO_2 is:

- $2\text{SrO}_2 \rightarrow 2\text{SrO} + \text{O}_2$; [1 mark]

ii) The temperature at which thermal decomposition of MO_2 occurs varies down Group 2 by:

- Temperature will increase (down the group); [1 mark]

- (Because,) the charge density of the cation decreases (down the group); [1 mark]
- This means less polarisation of the O_2^{2-} ion or weakens the O-O bond less; [1 mark]

[Total: 3 marks]

- The question gives you a clue
 - You are told that the thermal decomposition of MO_2 compounds is similar to that of the decomposition of Group 2 metal carbonates
 - $\text{MCO}_3 \rightarrow \text{MO} + \text{CO}_2$
 - In this case, there is no carbon atom present to form carbon dioxide as a product
 - The clue is actually relating to the formation of a **solid** product and a **gaseous** product
 - $\text{SrO}_2 \rightarrow \text{SrO} + \text{O}_2$
 - SrO is the solid and O_2 is the gaseous product
- The trend in the temperature required for thermal decomposition to occur increases down the group for the same reason as for carbonates and nitrates

2a

a) The factors that are responsible for the trend in solubility of the Group 2 hydroxides are:

- Lattice energy
AND
Hydration energy; [1 mark]

Explanation:

- Lattice energy and hydration energy both decrease
AND
Lattice energy decreases more than hydration energy; [1 mark]
- Enthalpy change of solution becomes more negative / exothermic
AND
hydroxides become more soluble; [1 mark]

[Total: 3 marks]

- This is a common question in Group 2 chemistry so make sure you learn the

marking points

- For small negative ions like hydroxide, the lattice enthalpy falls faster than the hydration enthalpy of the positive ions
 - That means that the enthalpy of solution will become less positive (or more negative)
- This is the opposite for larger negative ions like sulfate or carbonate
 - For these, the hydration enthalpy of the positive ions falls faster than the lattice enthalpy and so the sulfates get less soluble.
- That means the enthalpy of solution will become more positive (or less negative)

2b

b) $\text{Mg}(\text{OH})_2$ cannot be used to test for carbon dioxide, but $\text{Ba}(\text{OH})_2$ can because:

- $\text{Mg}(\text{OH})_2$ is not very soluble; [1 mark]
- $\text{Ba}(\text{OH})_2$ is soluble

OR

BaCO_3 is insoluble; [1 mark]

[Total: 2 marks]

- The reaction of $\text{Ba}(\text{OH})_2$ with CO_2 is similar to the reaction of $\text{Ca}(\text{OH})_2$ and CO_2
 - $\text{Ba}(\text{OH})_2 (\text{aq}) + \text{CO}_2 (\text{g}) \rightarrow \text{BaCO}_3 (\text{s}) + \text{H}_2\text{O} (\text{l})$
- A hydroxide solution must be at least sparingly soluble for a precipitate to form when testing for carbon dioxide

2c

c)

i) The expression for K_{sp} is:

- $K_{\text{sp}} = [\text{Sr}^{2+}][\text{OH}^-]^2$; [1 mark]

ii) The value for K_{sp} is:

- $[\text{Sr}(\text{OH})_2] = \frac{17.70}{121.6} = 0.146 \text{ mol dm}^{-3}$; [1 mark]
- $K_{\text{sp}} = (0.146) \times (2 \times 0.146)^2 = 0.012$; [1 mark]
- units = $\text{mol}^3 \text{ dm}^{-9}$; [1 mark]

[Total: 3 marks]

- **Careful:**

- The data in Table 3.1 is given in g dm^{-3} , so you must convert to mol dm^{-3} for this calculation
- For every Sr^{2+} ion, there are **two** OH^- ions
 - This means that you have to double the concentration of Sr^{2+} to get the correct concentration of OH^-
 - This step is often missed out by students
- The K_{sp} expression has $[\text{OH}^-]^2$
 - For the calculation, this means that you need to double the concentration of Sr^{2+} **AND** work with the $[\text{OH}^-]^2$ term
 - A common mistake made by students is to either double the concentration **OR** work with the $[\text{OH}^-]^2$ term

2d

d) To calculate the pH:

correct concentration of OH^-

- This step is often missed out by students
- The K_{sp} expression has $[\text{OH}^-]^2$
 - For the calculation, this means that you need to double the concentration of Sr^{2+} **AND** work with the $[\text{OH}^-]^2$ term
 - A common mistake made by students is to either double the concentration **OR** work with the $[\text{OH}^-]^2$ term

2d

d) To calculate the pH:

- Concentration of $\text{Mg}(\text{OH})_2 = \frac{0.140}{58.3} = 0.0024 \text{ (mol dm}^{-3}\text{)}; [1 \text{ mark}]$
- $\text{Mg}(\text{OH})_2 \rightarrow \text{Mg}^{2+} + 2\text{OH}^-; [1 \text{ mark}]$
- $[\text{OH}^-] = 2 \times 0.0024 = 0.0048 \text{ mol dm}^{-3}; [1 \text{ mark}]$
- $K_{\text{w}} = [\text{OH}^-][\text{H}^+]$
- $[\text{H}^+] = \frac{0.681 \times 10^{-14}}{0.0048} = 1.42 \times 10^{-12}; [1 \text{ mark}]$
- $\text{pH} = 11.8 / 11.85; [1 \text{ mark}]$

[Total: 5 marks]

- Many students will miss the step of multiplying the concentration of $\text{Mg}(\text{OH})_2$ by two in order to find the concentration of $[\text{OH}^-]$ ions
 - $\text{Mg}(\text{OH})_2 \rightarrow \text{Mg}^{2+} + 2\text{OH}^-$
- Once you have this, you can then use the K_{w} expression to calculate $[\text{H}^+]$
 - $K_{\text{w}} = [\text{OH}^-][\text{H}^+]$
 - $[\text{H}^+] = \frac{K_{\text{w}}}{[\text{OH}^-]}$
- As the temperature is 293 K and not 298 K you must use the K_{w} expression and not 14 - (-log $[\text{H}^+]$)

3a

a)

i) To show calcium carbonate is stable at 25 °C:

- $\Delta S = 40 + 214 - 89 = 165 \text{ (J K}^{-1}\text{mol}^{-1}\text{)}; [1 \text{ mark}]$

- $\Delta G = +178 - (298 \times 0.165)$; [1 mark]
- $\Delta G = (+) 129 \text{ kJ mol}^{-1}$; [1 mark]
- ΔG is above 0 so the reaction is not feasible

OR

As ΔG is above 0, CaCO_3 is stable; [1 mark]

ii) To calculate the minimum temperature needed for calcium carbonate to decompose:

- $0 = \Delta H - T\Delta S$

OR

$$\Delta H = T\Delta S$$

OR

$$T = \Delta H \div \Delta S; [1 \text{ mark}]$$

- $178 \div 0.165 = 1079 \text{ K}$

OR

806°C ; [1 mark]

[Total: 6 marks]

- The first part of the question is basically asking you to show that the thermal decomposition of calcium carbonate is **not** feasible at 25°C
- To determine this use, $\Delta G = \Delta H - T\Delta S$
 - You are given ΔH in the question and need to calculate ΔS using:
 - $\Delta S^\ominus = \sum S_{\text{products}}^\ominus - \sum S_{\text{reactants}}^\ominus$
- Once you have calculated ΔS :
 - Your answer of $165 \text{ J K}^{-1} \text{ mol}^{-1}$ needs converting to $\text{kJ K}^{-1} \text{ mol}^{-1}$ by dividing by 1000
 - You also need to convert 25°C into K by adding 273
- **Remember:** If ΔG is above 0 the reaction is not feasible
 - In this case, ΔG is 129 kJ mol^{-1} so the calcium carbonate will not decompose and is therefore stable
- Using your figures from part i), you can calculate the temperature at which the reaction becomes feasible
 - You assume that $\Delta G = 0$, rearrange and evaluate the equation to find T
- You **must** specify units in your answer to achieve the final mark

3b

b) The trend in the thermal decomposition of the Group 2 carbonates is:

- (Down the group) carbonates become more stable

OR

(Down the group) carbonates are more difficult to decompose

OR

(Down the group) carbonates need a higher temperature to decompose; [1 mark]

- Because there is less polarisation of the anion / nitrate ion / C-O bonds [1 mark]
- As the radius of M^{2+} / metal ion increases

OR

As the charge density of the cation decreases; [1 mark]

[Total: 3 marks]

- Describing and explaining trends in thermal decomposition of the Group 2 nitrates and carbonates is a common question, so take time to learn the marking points
- Apart from the difference in the bond (N-O and C-O) the marking points are the same for both the thermal decomposition of Group 2 nitrates and carbonates

3c

c)

i) The equation is:

- $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{Ca}^{2+}(\text{aq}) + \text{CO}_3^{2-}(\text{aq})$; [1 mark]

ii) If a few drops of concentrated K_2CO_3 are added:

- (White) solid appears / precipitation (of CaCO_3); [1 mark]
- As $[\text{CO}_3^{2-}]$ increases shifting equilibrium to the LHS (precipitating out CaCO_3); [1 mark]

[Total: 3 marks]

- This question is asking about the common ion effect in the context of Group 2 metals
- If a common ion, in this case CO_3^{2-} , is introduced to a saturated solution of an ionic compound, the equilibrium will shift

- This causes a precipitate to form as by adding the CO_3^{2-} ions the solubility of CaCO_3 is reduced

3d

d) To calculate the solubility:

- Solubility = $\sqrt{5 \times 10^{-9}} = 7.07 \times 10^{-5} \text{ mol dm}^{-3}$; [1 mark]
- Solubility = $7.07 \times 10^{-5} \times 100.1 = 7.08 \times 10^{-3} \text{ g dm}^{-3}$; [1 mark]

[Total: 2 marks]

- **Remember:** Watch out for units
 - K_{sp} expressions contain the concentrations of the ions, not the mass, so you must convert mol dm^{-3} to g dm^{-3} for this calculation as you are asked to express your answer in g dm^{-3}

